

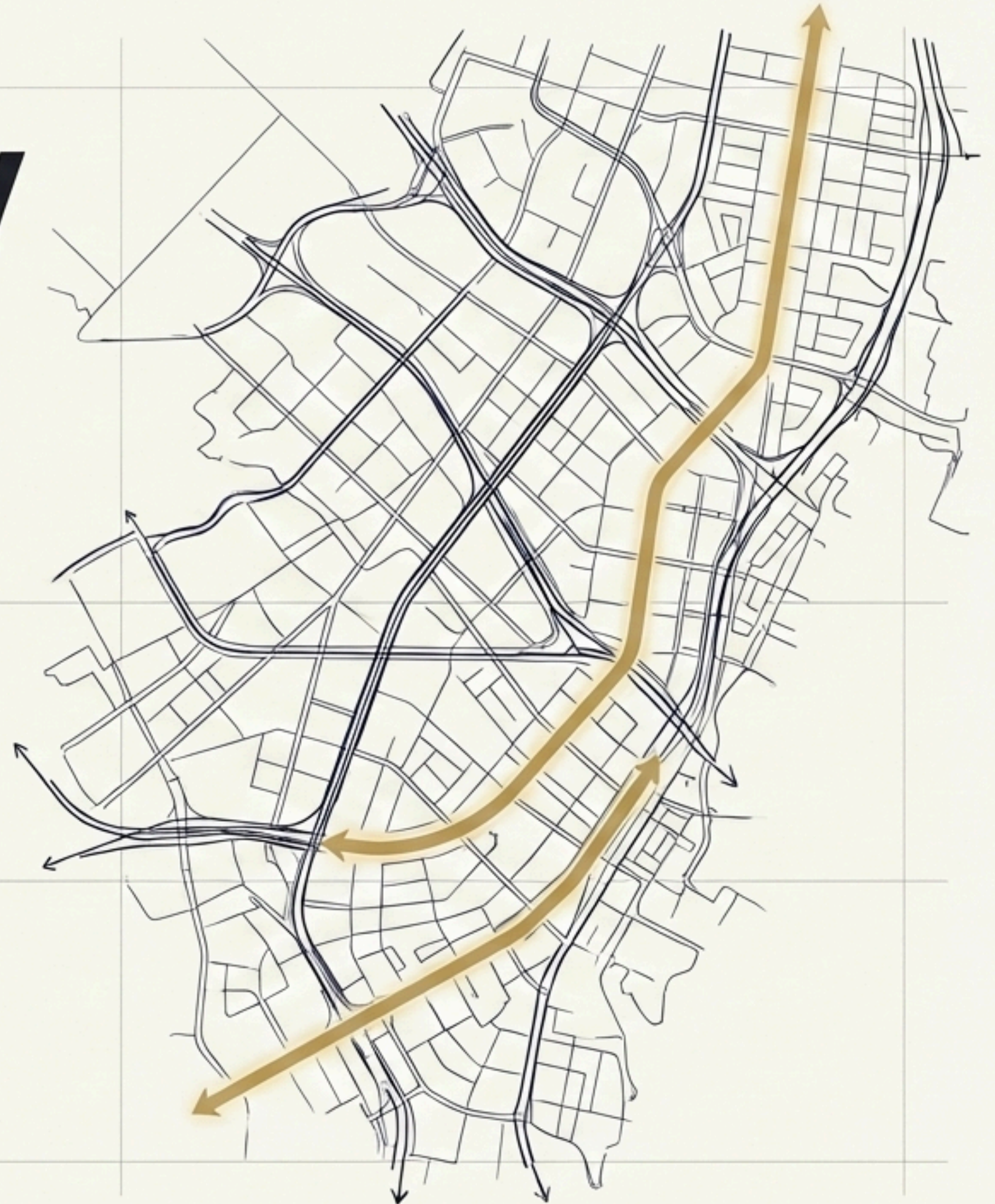
Urban Mobility Optimization System

A Systems Analysis and Design
Approach for University Transport
in Bogotá

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The Baseline Human Experience

312

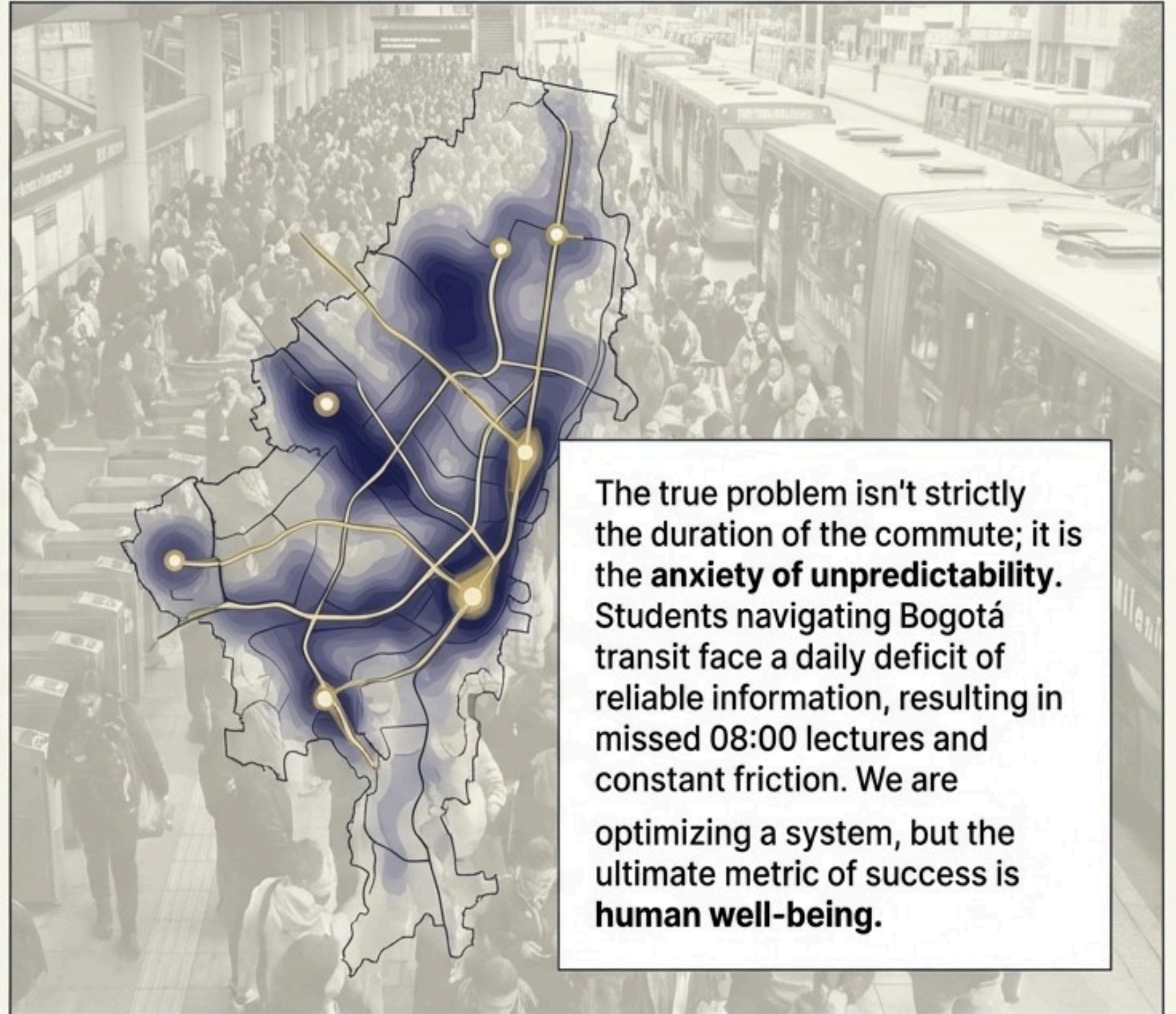
Commuters Surveyed

67 Minutes

Average one-way commute

82 Minutes

Multimodal TM + SITP commuters

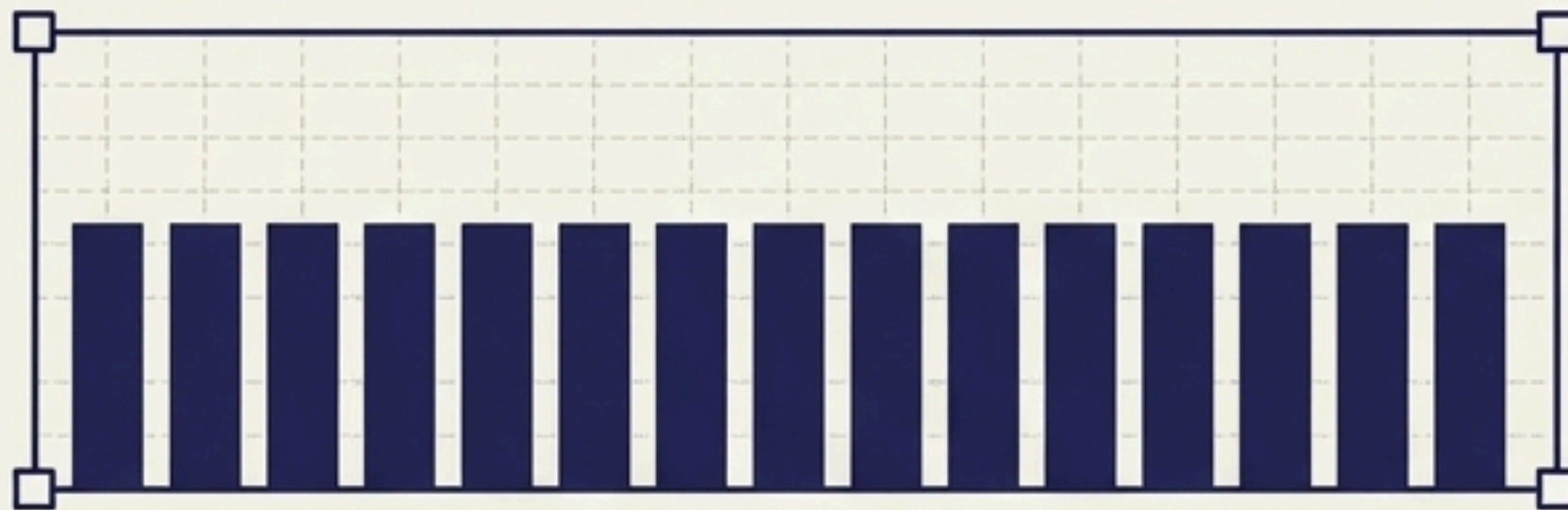


The true problem isn't strictly the duration of the commute; it is the **anxiety of unpredictability**. Students navigating Bogotá transit face a daily deficit of reliable information, resulting in missed 08:00 lectures and constant friction. We are optimizing a system, but the ultimate metric of success is **human well-being**.

The Catalyst of Chaos

The transport network functions adequately under ideal conditions, but fractures completely under environmental stress.

Deterministic State (Dry Day)



CONSISTENT ARRIVALS



Average Wait Time: **19 minutes**

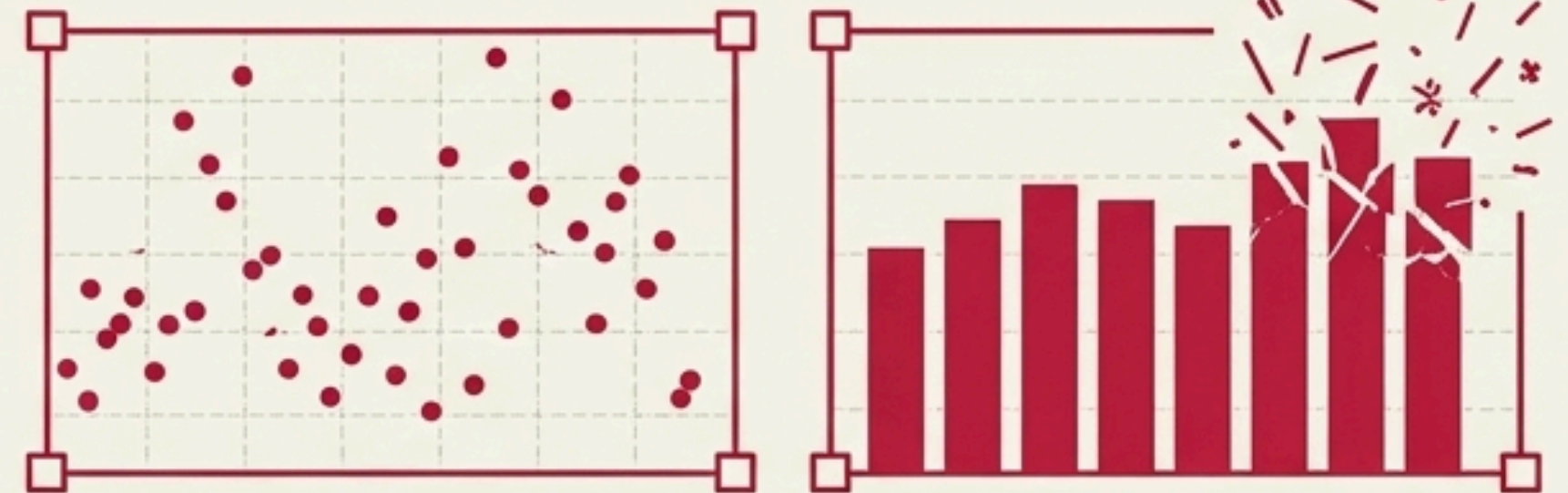


SITP Occupancy: **74%**



Bicycle Arrivals: **62**

Stochastic State (Rainy Day)



ERRATIC ARRIVALS

SYSTEM BREAKDOWN & OVERFLOW



Average Wait Time: **31 minutes (+63.2%)**



SITP Occupancy: **108%** (Critical overflow)



Bicycle Arrivals: **4 (-93.5% collapse)**

Reframing the Problem

We began this project intending to build a standard multimodal routing app. However, our analysis revealed that the transport network around UDFJC defies standard software engineering assumptions.

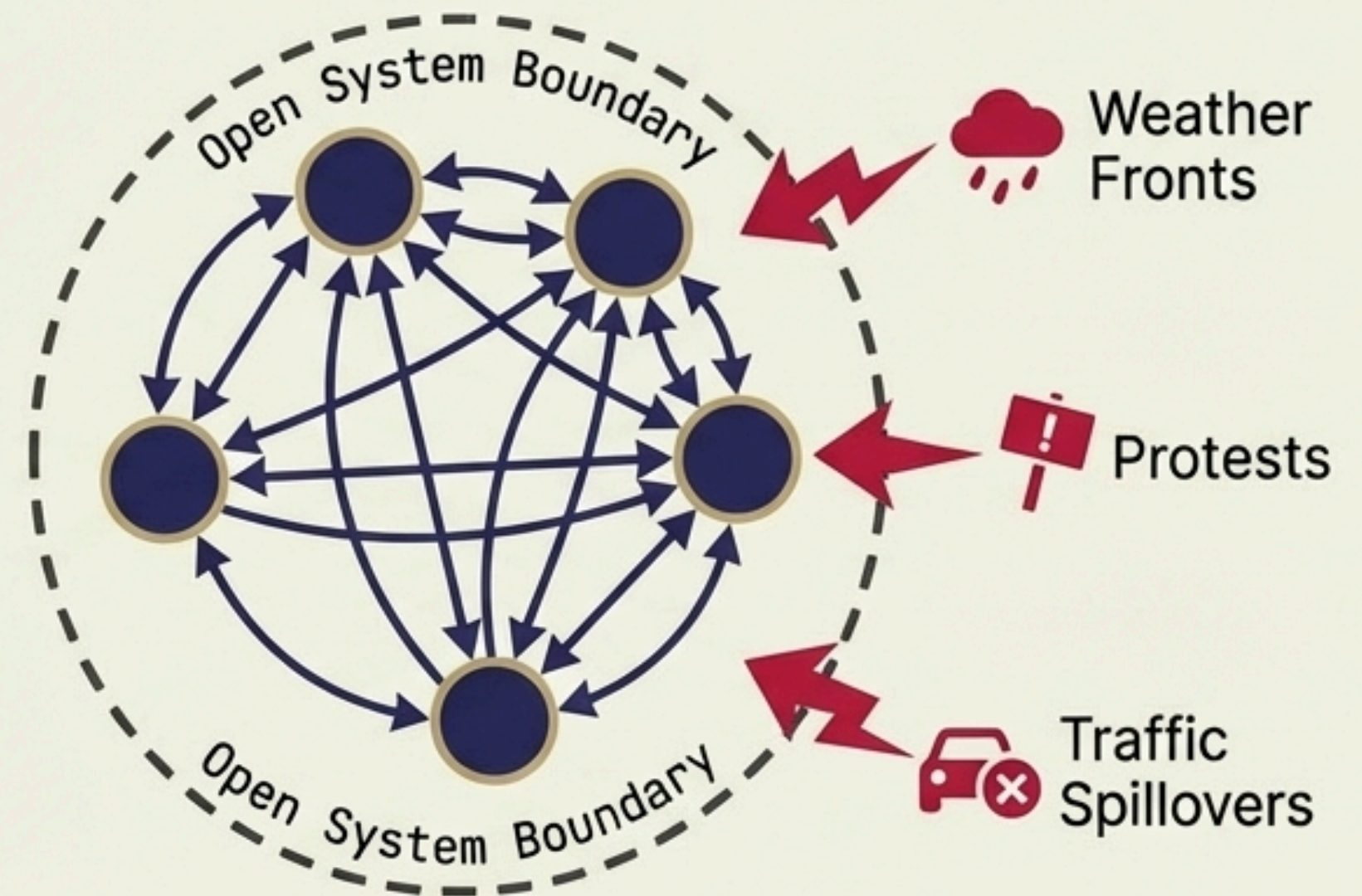
UMOS Paradigm: Complex Adaptive System

The system is not merely complicated; it is complex. It exchanges matter and energy with its environment. Standard deterministic algorithms cannot optimize an open system.

Standard Approach



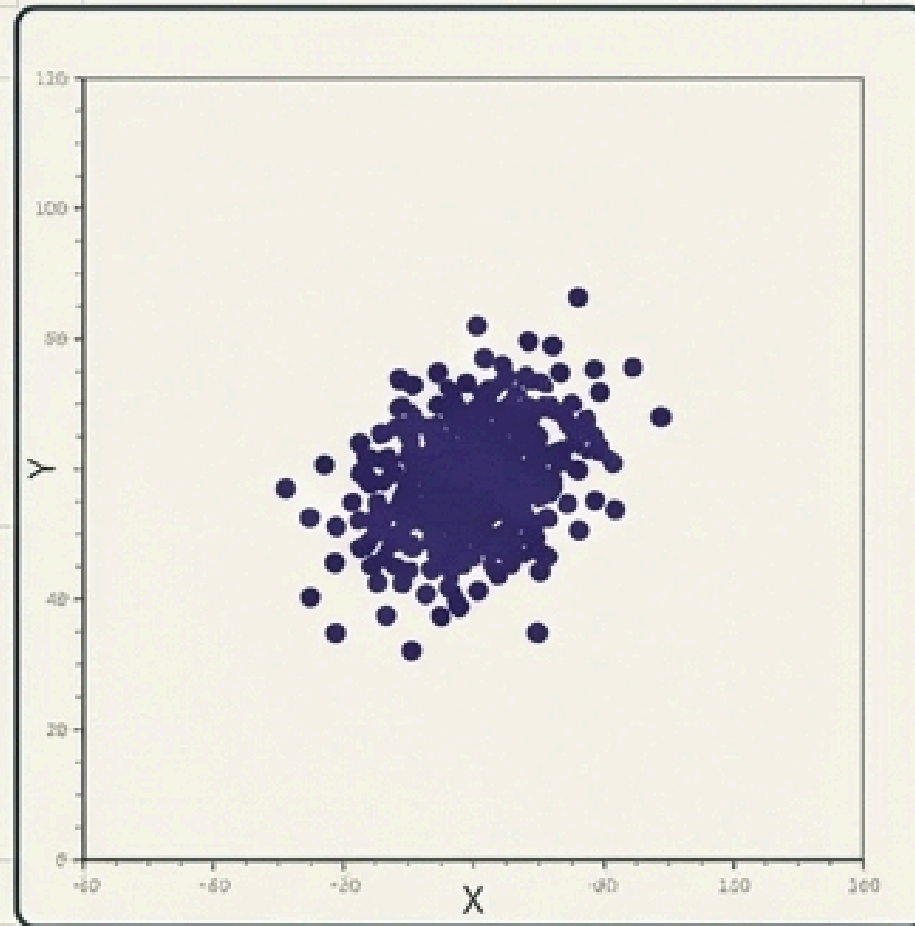
The network is complicated, but static.



Quantifying Unpredictability

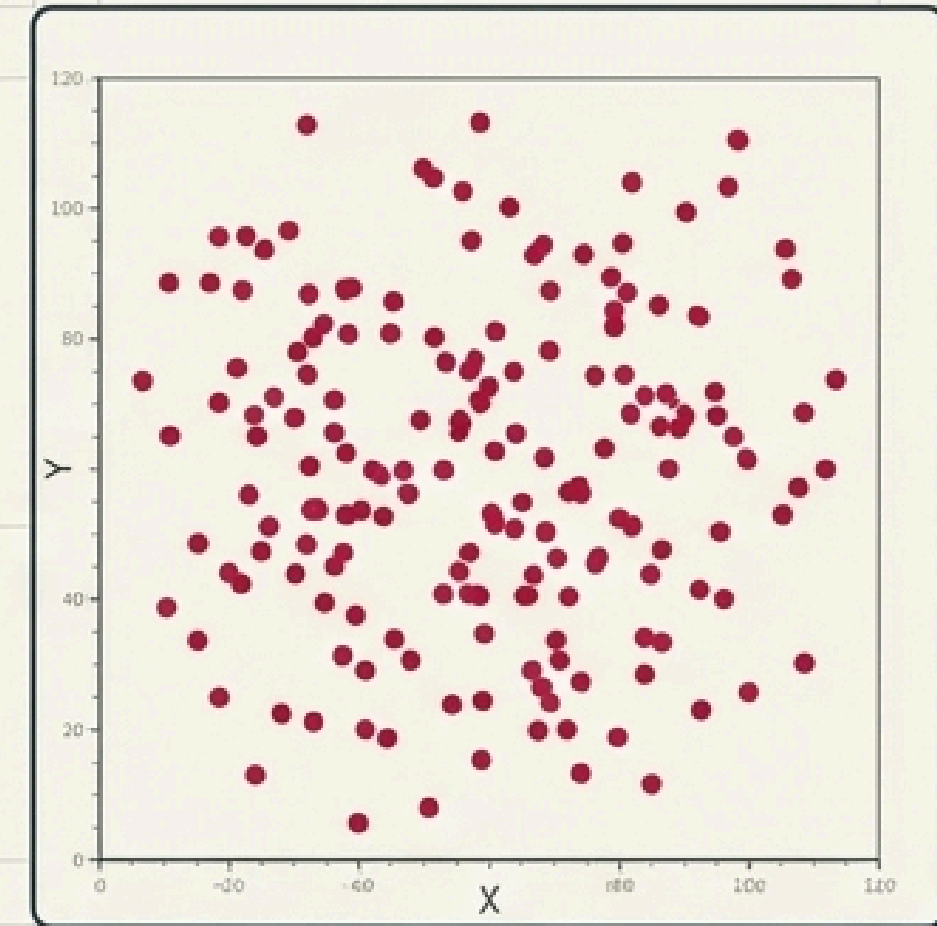
Applying Shannon's Entropy to the commute. Rain causes a 128% jump in relative variability. Wait times scatter randomly.

The Entropy Gauge



Low Entropy (CV = 0.18)

Timely Information



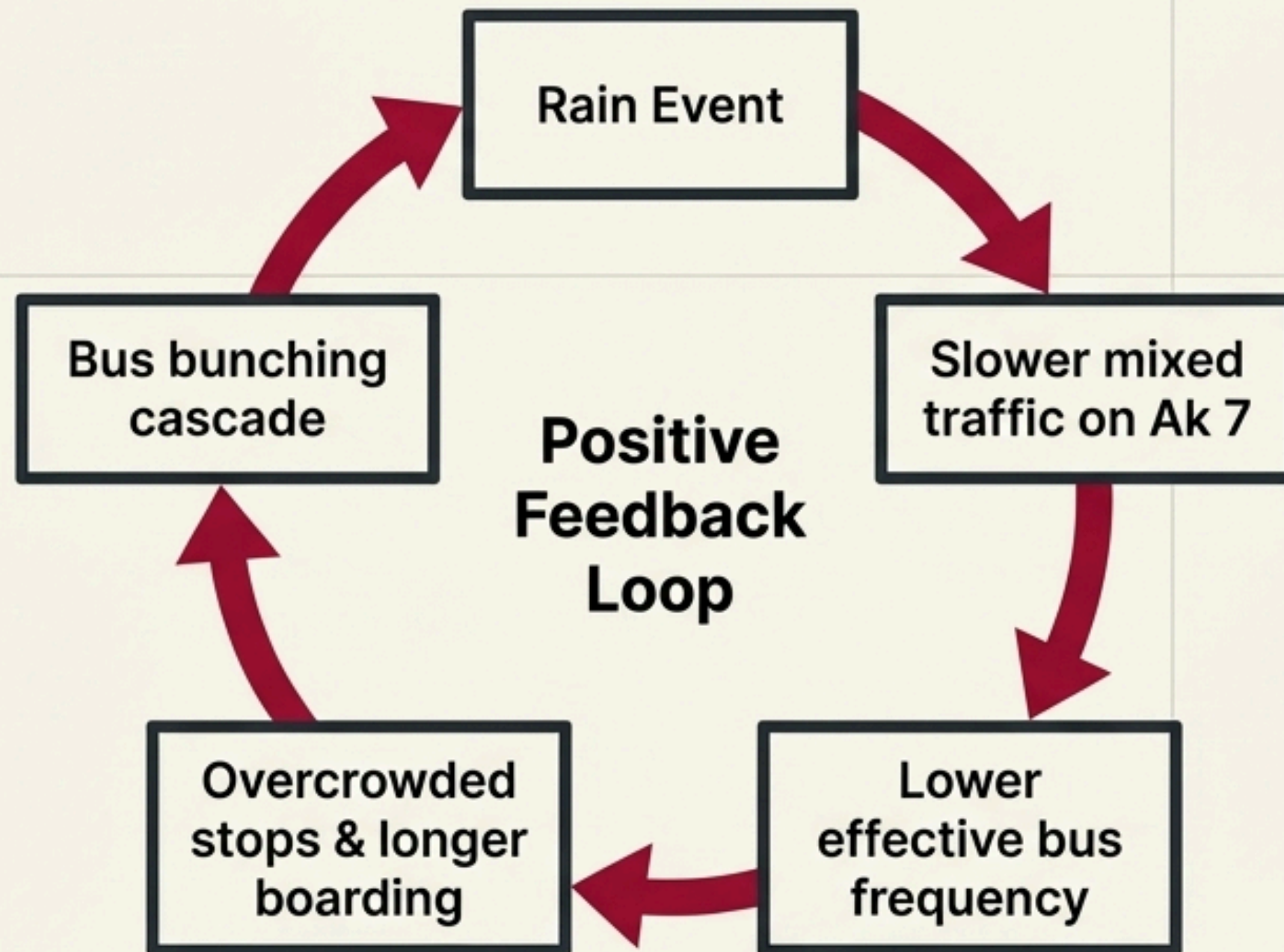
High Entropy (CV = 0.41)

The Engineering Response:

UMOS cannot fix the weather, but information is the classical antidote to entropy. Because 39% of students distrust existing apps, UMOS enforces a strict UI requirement: every single data point must display its precise age 🕒 (e.g., "Updated 45s ago") to combat distrust in a high-entropy environment.

Non-Linear Cascading Failures

Our Workshop #1 sensitivity analysis revealed the network's behavior is dominated by a handful of parameters that interact non-linearly.

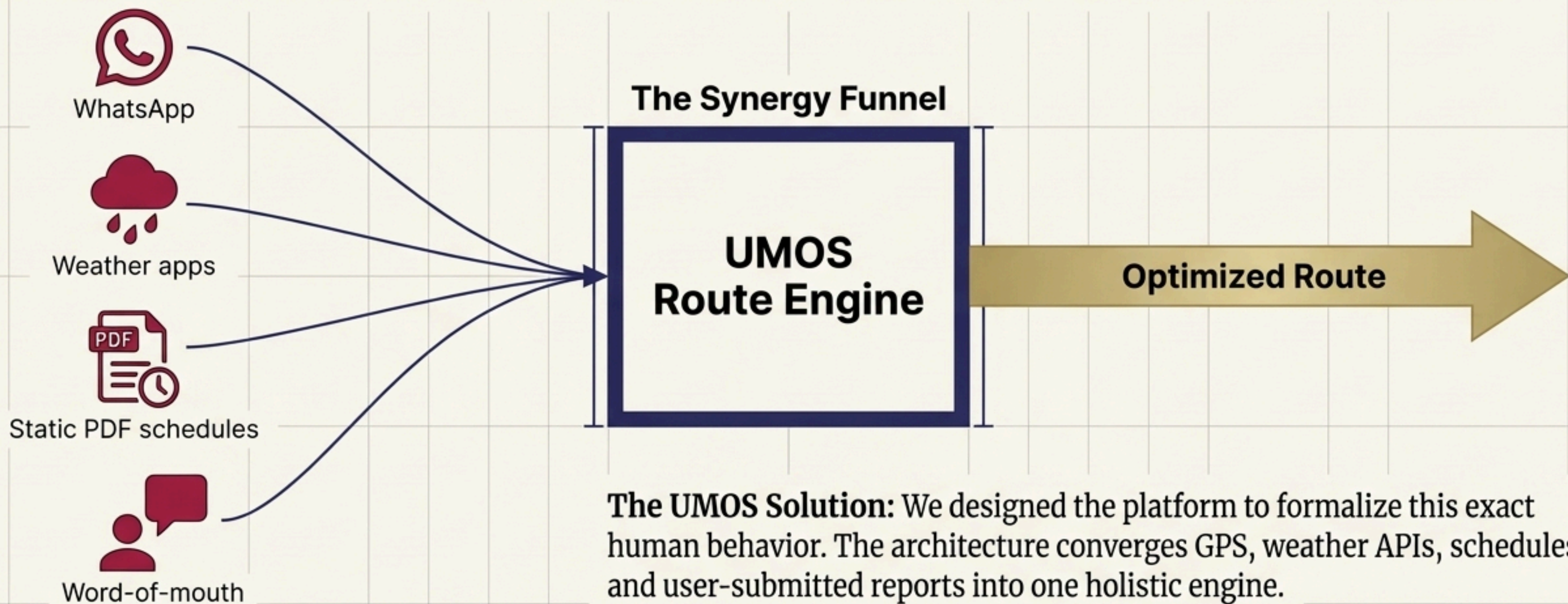


SENSITIVITY DATA

A 10% increase in rainfall intensity does not yield a 10% increase in wait times. It triggers a **26.4% exponential increase** due to structural bottlenecks (Estación Universidades single hub) and boarding delays.

Formalizing Emergent Synergy

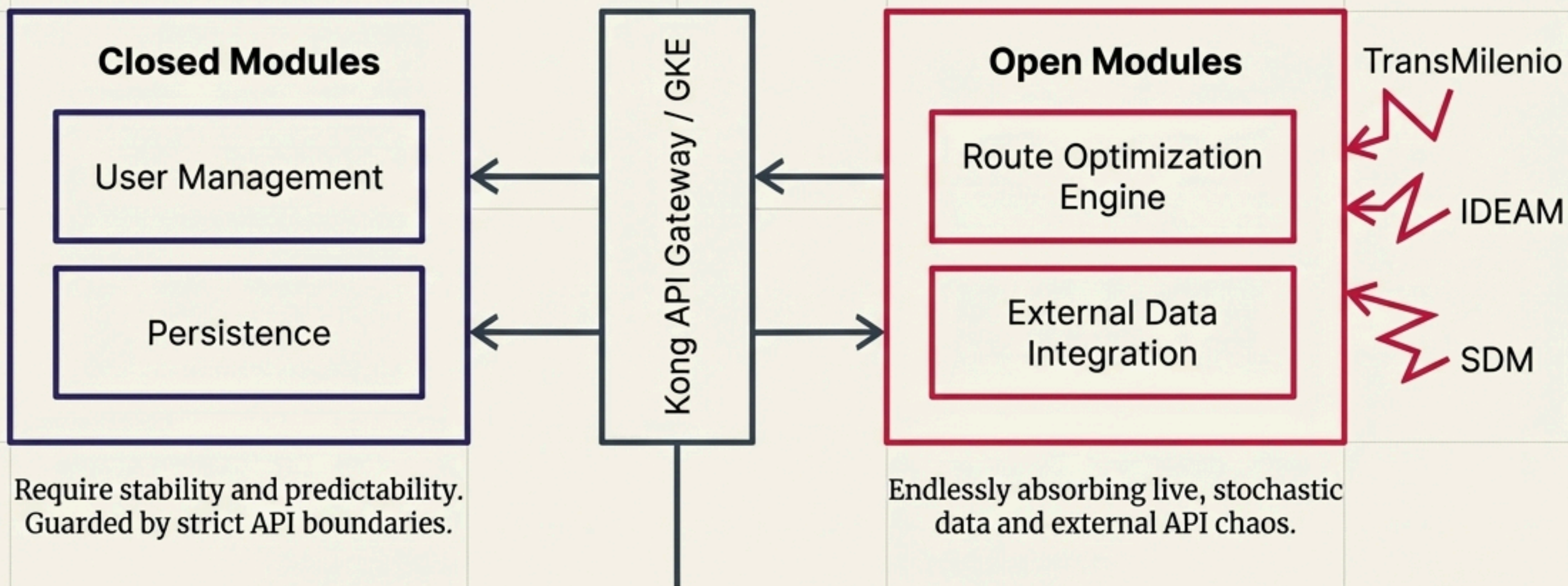
Students had already built an informal optimization system: WhatsApp groups crowd-sourcing arrival times and station crowds. This is **emergent synergy**—the collective intelligence outperforming official sources.



The UMOS Solution: We designed the platform to formalize this exact human behavior. The architecture converges GPS, weather APIs, schedules, and user-submitted reports into one holistic engine.

Architectural Tension

Built on Node.js/NestJS and React Native. We **deliberately maintain** a tension between modules based on their entropy exposure.



Kong Gateway protects system stability from external API failures, isolating components based on their environmental risk.

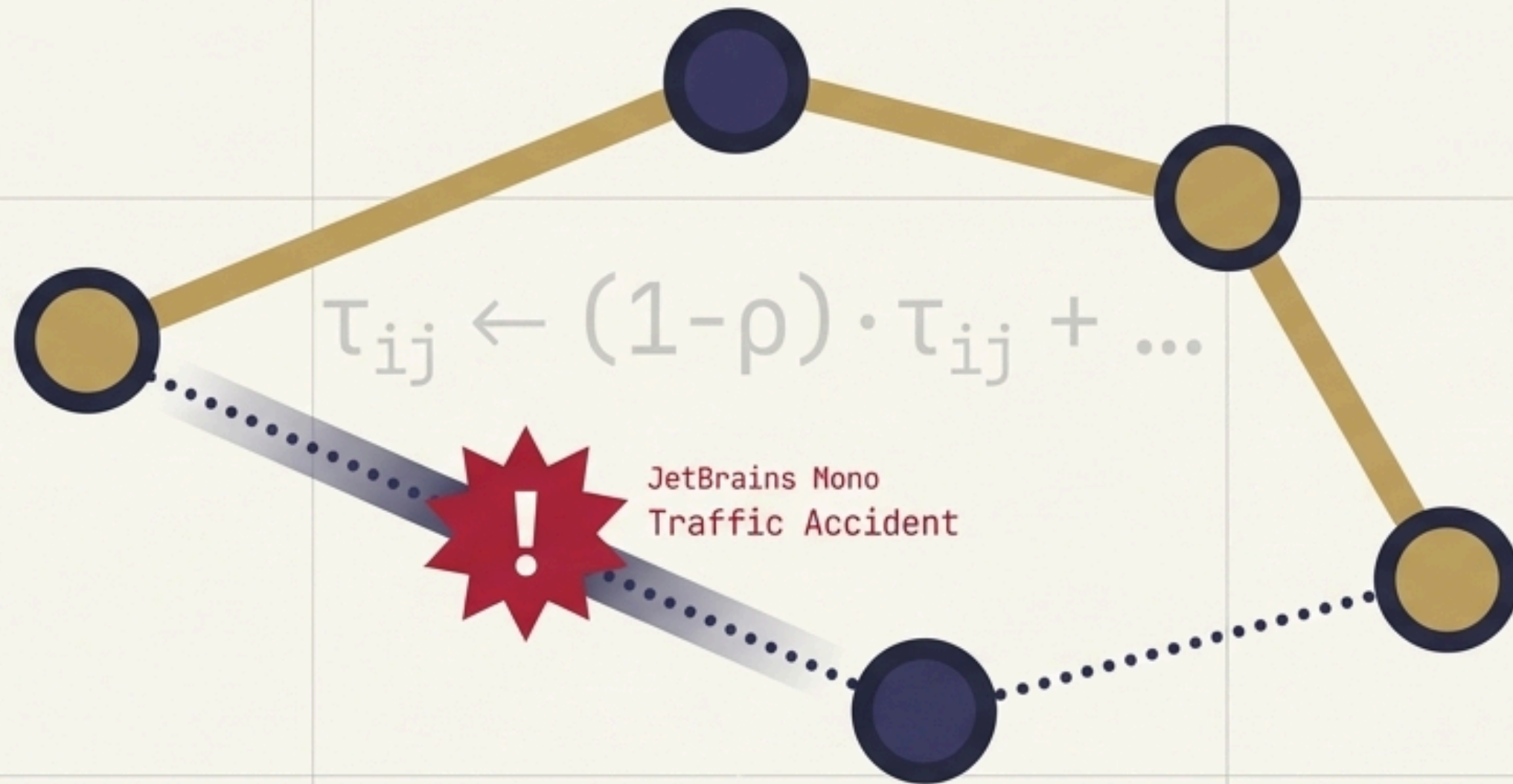
The Algorithmic Dilemma

	Dijkstra's Algorithm (Classical Approach)	Ant Colony Optimization (Biological/Stochastic Approach)
Edge Weights	Fixed static numbers.	Dynamic probabilities (τ and η).
Network State	Assumes a perfectly static graph.	Handles a live, mutating 180-node / 420-edge graph. JetBrains Mono
Reaction to Rain	Computes yesterday's optimal route.	Adapts in real-time to current physical conditions.
Verdict	Mathematically Insufficient	Structurally Required

Biologically Inspired Routing

UMOS deploys 50 digital ants per query across 20 iterations, converging in under 800ms.

The Pheromone Loop



The Power of Forgetting

The most critical variable is the evaporation rate ($\rho = 0.15$).

In a stochastic system, if a route was great yesterday but is jammed today, stale data is actively harmful.

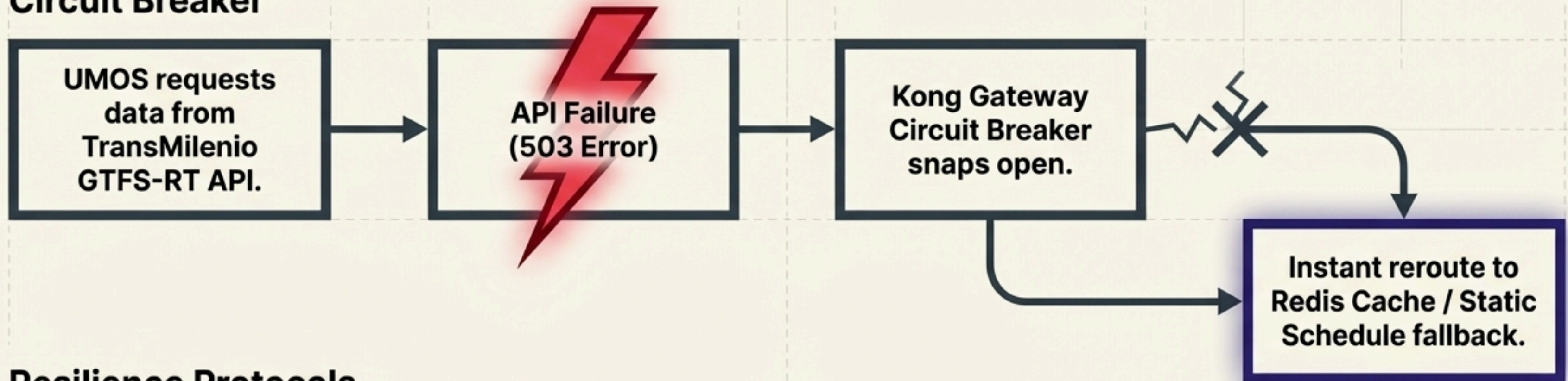
Actively "forgetting" 15% of historical data each cycle ensures the system only trusts live realities.

JetBrains Mono
Convergence: <800ms

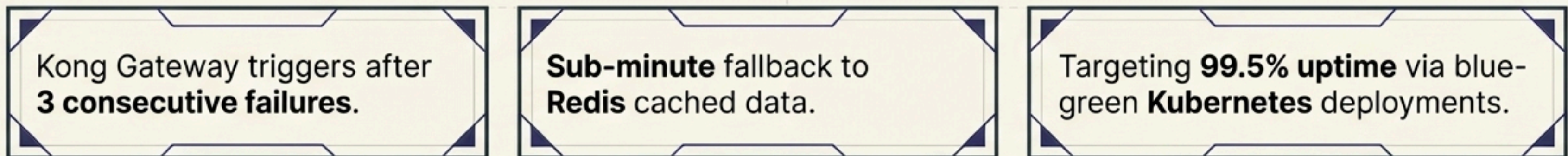
Engineering for Inevitable Failure

Academic projects often assume perfect conditions. UMOS is designed for the reality of Bogotá's digital infrastructure.

Circuit Breaker

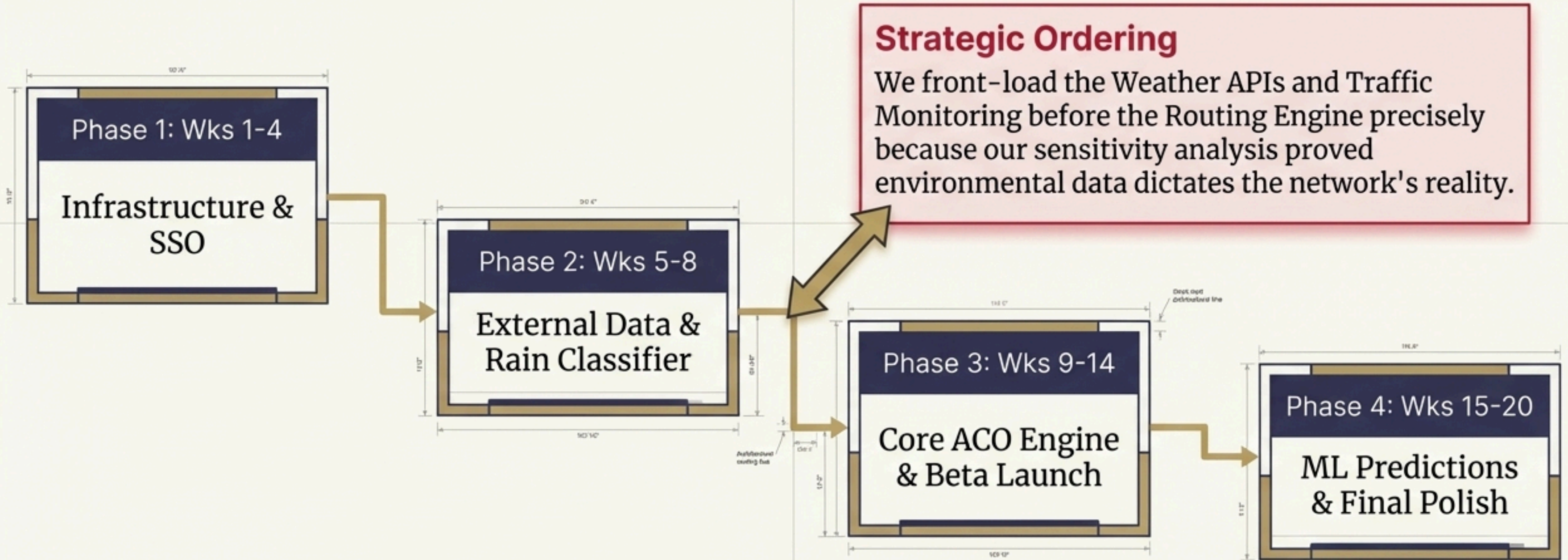


Resilience Protocols



Deployment Pipeline

A 20-week, 10-sprint path to a working Semester 2026-I prototype.



Restoring Peace of Mind



We are not going to fix Bogotá's traffic. That requires systemic infrastructure changes far beyond the scope of a software project.

The True Impact:

If UMOs can shave 9 minutes off a commute, prevent a student from missing their 08:00 lecture, and replace the daily anxiety of the unknown with mathematical confidence, we have succeeded in our engineering mandate.